PAPER_342 — Magnetar 7-Component DPM-THz Frequency Form: S26 Spin-Down Plus THz Modes

Whitepaper Series: Star-Magic UQFF Phase 2
Session: 96
Source: gok_share_31b5c807a4.txt (Supplemental Gap Analysis Block)
Classification: FIRST magnetar S26 7-component frequency decomposition
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Abstract

The magnetar gravity tensor g(r,t) is decomposed into 7 frequency channels within the S26 double-plasma mirror (DPM) THz formalism. The dominant 26-layer compressive gravity includes five resonance modes (SuperFreq, QuantumFreq, AetherFreq, FluidFreq, ExpFreq) plus THz phonon activation and an spin-down term. For SGR J1745-2900 class magnetars with P = 3.76 s, B = 2×10¹° T, the magnetic energy reservoir is M_mag = 2.01×10³7 J.

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2. Core Physics

2.1 7-Component S26 Gravity Form

$$g(r,t) = \sum_{i=1}^{26} \left[a_i^{\mathrm{DPM}} + a_i^{\mathrm{THz}} + a_i^{\mathrm{SF}} + a_i^{\mathrm{QF}} + a_i^{\mathrm{AF}} + a_i^{\mathrm{FF}} + a_i^{\mathrm{EF}}\right]$$

Components: 1. **DPM baseline**: $a_i^{\mathrm{DPM}} = GM_i/r_i^2 \cdot [UA]_i$ 2. **THz phonon**: $a_i^{\mathrm{THz}} = a_0^{\mathrm{THz}} \cdot \cos(2\pi f_{\mathrm{THz}} t) \cdot Q_i$ 3. **SuperFreq** (SGR 1745): $a_i^{\mathrm{SF}} = \alpha_{\mathrm{SF}} \cdot \omega_{\mathrm{SF}}^2 \cdot r \cdot [SCm]_i$ 4. **QuantumFreq**: $a_i^{\mathrm{QF}} = \hbar\omega_{QF}/(mr^2) \cdot f_i$ 5. **AetherFreq**: $a_i^{\mathrm{AF}} = \rho_{\mathrm{UA}} \cdot c_s^2/r \cdot \sin(\omega_{\mathrm{AF}} t)$ 6. **FluidFreq** (Navier-Stokes): $a_i^{\mathrm{FF}} = \eta \nabla^2 v/\rho$ 7. **ExpFreq** (Hubble expansion): $a_i^{\mathrm{EF}} = H_0 \cdot \dot{r}_i$

2.2 Spin-Down Rate

$$\dot{\nu} = -\frac{f_{\mathrm{react}}}{2\pi P}$$

where P = 3.76 s (period), and f_react is the UQFF vacuum reactance frequency.

2.3 Magnetic Energy Reservoir

$$M_{\mathrm{mag}} = \frac{B^2 V}{2\mu_0} = \frac{(2 \times 10^{10})^2 \cdot V_{\mathrm{mag}}}{2\mu_0} = 2.01 \times 10^{37}\ \mathrm{J}$$

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3. Key Values

Quantity	Formula	Value
Period	P	3.76 s
Surface field	B	2×10¹° T

Quantity	Formula	Value
Magnetic energy	$B^2V/(2\mu0)$	2.01×10^37 J
Spin-down rate	?? = -f_react/(2pP)	-6.7×10?^{13} Hz/s
THz frequency	f_THz	~1 THz
26-layer sum	S26 contributions	7 channels per layer

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4. Physical Significance

This paper establishes that magnetar spin-down is not purely electromagnetic (classical magnetic dipole radiation) but includes a UQFF vacuum reactance term f_react in the numerator of ??. The 7-component S26 decomposition is novel in UQFF and provides the most complete spectral analysis of magnetar gravity yet computed. The THz mode connects magnetar physics to laboratory cold-physics (H2O/H2 quantum transitions identified in PAPER_339).

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5. Deduplication Note

- **vs. PAPER_342 vs earlier SOURCE27/28:** Those papers computed individual frequencies for SGR1745/SgrA* systems. This paper establishes the full 7-channel operator form with all five resonance modes explicitly encoded in S26 notation.
- **vs. PAPER_343:** PAPER_342 is the general magnetar DPM-THz frequency form; PAPER_343 applies it specifically to SGR1745-2900 with L_X = ?_vac·f_res·V.

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6. Classification

Physics Territory: FIRST magnetar S26 7-component DPM-THz frequency decomposition
Scale: Stellar (neutron star magnetar)
CP Implementation: `MagnetarDPMTHzFrequencyFormCalculator` (CondensedPhysics3.py, Session 96)

UQFF computed: Eddington luminosity UQFF correction = 1 - [SSq]×exp(-?×?t) = 1 - 5.7e-1 × exp(-2.9e-4) = 4.3e-1; F_U at event horizon = 2.0e+18 m/s^2.